

DataBridge

by:



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Gitrepo Link:

<https://github.com/AlecHwvu/CapstoneDataBridgeFileMaint.git>

I. ABSTRACT

1.1 Summary

Rural hospitals face challenges associated with legacy electronic health (EHR) systems, interoperability, and inefficient documentation workflows. Recent research has indicated that while EHR adoption has become widespread across the nation's hospitals, there is a notable fall-behind in rural hospitals in relation to advanced EHR capabilities, primarily interoperability and data sharing. This project proposes a browser-based bedside documentation system that is designed to demonstrate how a modern, tablet-friendly interface can collect patient information, and transmit standardized data to an EHR intake service.

1.2 Key Concepts

Ideally, the project will combine several computer science and healthcare information technology concepts, including responsive web development, application programming interfaces (APIs), authentication, role-based access control (RBAC), databases, audit logging, optical character recognition (OCR), document classification, encryption, and healthcare interoperability. [HL7 identifies Version 2 and FHIR among its widely implemented healthcare interoperability standards.](#)¹ This project will primarily investigate HL7 Version 2 for its simulated EHR communication component while using Python and FastAPI for the backend, with React or standard HTML/CSS/Java considered for the frontend.

1.3 Problem Statement

A lot of healthcare organizations have to maintain and exchange patient information while operating with systems of different capabilities and ages. Rural hospitals likely are presented with additional challenges because of limited resources, and the ever-growing gaps in advanced EHR functionality. Legacy-system migration can also require organizations to manage information across both older and newer systems. This will be explained in greater detail in section 2.

1.4 Proposed Solution

The solution is a browser-based application optimized for use on a tablet at a patient's bedside. A nurse will authenticate into the system, select a patient from an active census, enter the appropriate observations/documentation, and submit the information. The backend will then validate the submission and convert applicable information into an HL7v2 message. From there, a simulated EHR receiver will receive and display the message, and demonstrate an end-to-end workflow without requiring access to a live hospital EHR. More details on the project and its features will be explained in greater detail in section three and four.

II. PROBLEM

2.1 The Real-World Problem

There is difficulty within the efficiency of documenting and transferring patient information in environments that rely on older or disconnected systems. There have been initial observations that indicate information is written down at the bedside before later being manually entered into

an EHR, which creates a plethora of opportunities for duplications, delayed documentation, and transcription errors. The intended workflow should be validated through interviews with those actively working within the healthcare system rather than assumption within every rural hospital. This project, therefore, will include working with nurses and other healthcare personnel to figure out how bedside documentation is performed, the inconveniences, which MEDITECH version or other EHR platform is being used, and what information would be the most useful to capture at the point of care.

2.2 Relevance

This problem is relevant considering it combines human-computer interaction, secure application development, databases, backend engineering, and healthcare interoperability into a single, real world system. The problem is also locally significant: rural hospitals in West Virginia counties like ours often operate on the same aging EHR platforms discussed in national research, meaning the documentation inefficiencies and interoperability gaps described in this proposal are not abstract, they affect the hospitals our own communities rely on. Addressing this problem at a local level gives the team direct access to real workflow input from healthcare staff, strengthening both the relevance and the practical grounding of the project.

III. BACKGROUND AND LITERATURE REVIEW

3.1 Existing Solutions

Rural hospitals continue to lag behind their urban counterparts in electronic health record (EHR) interoperability, a gap that recent national research confirms is both significant and ongoing

rather than resolving on its own. In response, EHR vendors such as MEDITECH have positioned platform modernization as their primary solution, actively encouraging rural hospitals still operating on legacy systems to migrate to newer, cloud based platforms like Expanse. Meanwhile the underlying standard that enables clinical data exchange across these systems, Health Level Seven Version 2 (HL7v2) remains the primary interoperability protocol that legacy hospital systems depend on, even as the broader healthcare industry gradually shifts its attention toward newer standards such as FHIR. Together, these existing efforts demonstrate that rural EHR interoperability is a recognized and active area of investment, though a meaningful gap remains for hospitals not yet positioned to undergo full platform migration.

3.2 Limitations

While these existing solutions show that rural EHR interoperability is an active area of investment, each leaves a gap this project addresses. The rural-urban interoperability gap is persistent rather than closing, with rural hospitals consistently 5-15 percentage points less likely than urban hospitals to query external records or integrate outside information into their EHR. This is largely attributed to rural hospitals' lower operating margins, limited capital, and smaller technical teams, which make the workflow redesign of advanced EHR functions difficult to sustain, pointing toward a need for lightweight low-cost tools rather than full system overhauls.

3.3 Solution

The proposed system will combine a tablet-optimized frontend, a Python/FastAPI backend, a database, an authentication and RBAC layer, an interoperability component, and a simulated EHR receiver. The nurse interface will allow a user to select a patient and enter the information at the bedside, and the backend will validate then process that information before

generating an HL7v2 message for the simulated receiver. With this, a separate administrative dashboard will display a unit-wide patient census and alerts, while an EHR receiver dashboard will display incoming messages. An audit-log interface will record who performed an action, and when. Additional organized tools will allow users to upload documents for classification and OCR-based metadata extraction without interpreting the medical content or providing diagnostic recommendations.

3.4 Justification

HL7 Version 2 was selected as this project's interoperability standard because it remains the format legacy MEDITECH systems are actually built to communicate in, despite its age.

Designing the interoperability layer around a newer standard such as FHIR, while a reasonable direction for future development, would target modern platforms that fall outside this project's rural, legacy-system focus. Choosing HL7v2 keeps the system grounded in the environments it is intended to serve, preserving the project's practical relevance.

VI. SYSTEM FEATURES

Primary Features

Primarily consists of the bedside documentation interface, user authentication, role-based access control, patient census selection, structured patient-data entry, backend validation, HL7v2 message generation, and the simulated EHR receiver. The system will provide separate views for nurses, charge nurses, and administrators, allowing the project to demonstrate that different users have different permissions. With the simulated EHR, displays of incoming HL7 messages occur

as theft is received, which provides a visible demonstration of the interoperability workflow from the bedside interface to the receiving system.

Secondary Features

Includes audit-log interface automatic organizational document classification, OCR-based extraction of metadata from scanned documents, secure medical-record transfer packages, and temporary one-time sharing links. The mentioned transfer packages will allow multiple documents to be grouped for a secure transfer, while sharing links will require authentication for designated recipients and expire after a defined period or number of uses. Document classification will be limited to organizational purposes; the goal of the system is to organize the data and documents, not diagnosing patients, interpreting medical findings, recommending treatments, or use as a clinical decision-making system. These features will be prioritized according to available development time after the primary bedside and interoperability workflow is functional.

V. PROJECT TIMELINE

Week	Dates	Task
1-3	08/18 - 09/08	Brainstorm and develop the project proposal, identify key concepts and unknown aspects, establish objectives and expected outcomes, and refine the concept through collaboration and peer feedback.
4-5	09/09 - 09/23	Develop and refine project requirements, conduct requirements gathering, research existing systems, and begin the Software Requirements Specification (SRS).
6-7	09/24 - 10/07	Continue developing the SRS, define functional and non-functional requirements according to IEEE standards, conduct peer review, and finalize the project's requirements and technical scope.
8-9	10/08 - 10/21	Complete and submit the SRS; begin initial system architecture, database design, frontend

		planning, authentication, and RBAC implementation.
10-11	10/22 - 11/04	Develop the bedside documentation interface, backend API, database functionality, validation, and initial HL7v2 interoperability proof of concept.
12-13	11/05 - 11/18	Finalize system architecture and HLD documentation, implement and test the simulated EHR receiver, and evaluate professional, ethical, societal, and sustainability considerations.
14-15	11/19 - 12/02	Continue system integration, testing, usability improvements, security features, audit logging, and secondary features as time permits.
16-17	12/03 - 12/15	Final testing, documentation, presentation preparation, system demonstration, project evaluation, and final report preparation.

VI. BUDGET & RESOURCES

Paid Item / Service	What we'd use it for	Estimated Cost
Auth0	Authentication, MFA, and RBAC	\$35
AWS	Backend/cloud hosting	\$60
Supabase Pro	Database, storage, and backups	\$25
Sentry	Application monitoring and security alerts	\$26
Cloudflare Pro	WAF, DDoS protection, HTTPS/security	\$25
1Password Business	Secure management of development credentials/secrets	\$8
Domain name	Professional project website	\$15
Security testing service/tools	Additional vulnerability/security testing	\$250
Cloud backup/storage	Additional backup space	\$20
Contingency	Unexpected expenses	\$Vary
TOTAL		≈ \$450-?